C++ ASSIGNMENT

TOPIC 1: CLASS

1. Student Class

```
#include<iostream>
using namespace std;
class Student
{
public:
string name;
int rollNo;
float marks;
void accept()
{
cin>>name>>rollNo>>marks;
}
void display()
{
cout<<"Name: "<<name<<endl;
cout<<"Roll No: "<<rollNo<<endl;
cout<<"Marks: "<<marks;
}
};
int main()
{
Student s;
s.accept();
s.display();
return 0;
}
```

Output

```
Amit 101 85

Name: Amit
Roll No: 101
Marks: 85
```

---

2. Circle Class

```
#include<iostream>
using namespace std;
class Circle
{
public:
float radius;
void calculate()
{
cout<<"Area: "<<3.14*radius*radius<<endl;
cout<<"Circumference: "<<2*3.14*radius;
}
};
int main()
{
```

```
Circle c;
c.radius=5;
c.calculate();
return 0;
}
```

Output

Area: 78.5

Circumference: 31.4

---

3. Product Class

```
#include<iostream>
using namespace std;
class Product
{
public:
int id,quantity;
float price;
void total()
{
cout<<"Total Price: "<<quantity*price;
}
};
int main()
{
Product p;
p.id=1;
p.quantity=5;
p.price=100;
p.total();
return 0;
}
```

Output

Total Price: 500

TOPIC 2: OBJECT

1. Rectangle Area and Perimeter

```
#include<iostream>
using namespace std;
class Rectangle
{
public:
int l,b;
void display()
{
cout<<"Area: "<<l*b<<endl;
cout<<"Perimeter: "<<2*(l+b);
}
};
int main()
{
Rectangle r;
r.l=10;
r.b=5;
r.display();
return 0;
}
```

Output

Area: 50

Perimeter: 30

---

2. Even or Odd

```
#include<iostream>
using namespace std;
class Number
{
public:
int n;
void check()
{
if(n%2==0)
cout<<"Even";
else
cout<<"Odd";
}
};
int main()
{
Number obj;
obj.n=8;
obj.check();
return 0;
}
```

Output

Even

---

3. Student Total and Percentage

```
#include<iostream>
using namespace std;
class Student
{
public:
int m1,m2,m3;
void result()
{
int total=m1+m2+m3;
float per=total/3.0;
cout<<"Total: "<<total<<endl;
cout<<"Percentage: "<<per;
}
};
int main()
{
Student s;
s.m1=80;
s.m2=75;
s.m3=85;
s.result();
return 0;
}
```

Output

Total: 240

Percentage: 80

TOPIC 3: cout <<

1. Display Personal Details

```
#include<iostream>
using namespace std;
int main()
{
cout<<"Name: Rahul"<<endl;
cout<<"Age: 18"<<endl;
cout<<"Department: CSE"<<endl;
cout<<"College: MIT ADT";
return 0;
}
```

Output

```
Name: Rahul
Age: 18
Department: CSE
College: MIT ADT
```

---

2. Multiplication Table

```
#include<iostream>
using namespace std;
int main()
{
int n=5;
for(int i=1;i<=10;i++)
cout<<n<<" x "<<i<<" = "<<n*i<<endl;
return 0;
}
```

Output

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
...
5 x 10 = 50
```

---

3. First 10 Even and Odd Numbers

```
#include<iostream>
using namespace std;
int main()
{
cout<<"Even Numbers"<<endl;
for(int i=1;i<=10;i++)
cout<<2i<<" ";
cout<<endl;
cout<<"Odd Numbers"<<endl;
for(int i=0;i<10;i++)
cout<<2i+1<<" ";
return 0;
}
```

Output

```
Even Numbers
2 4 6 8 10 12 14 16 18 20

Odd Numbers
1 3 5 7 9 11 13 15 17 19
```

TOPIC 4: cin >>

1. Accept Two Integers and Display Sum

```
#include<iostream>
using namespace std;
int main()
{
int a,b;
cin>>a>>b;
cout<<"Sum = "<<a+b;
return 0;
}
```

Output

10 20

Sum = 30

---

2. Area of Rectangle

```
#include<iostream>
using namespace std;
int main()
{
int l,b;
cin>>l>>b;
cout<<"Area = "<<l*b;
return 0;
}
```

Output

10 5

Area = 50

---

3. Positive, Negative or Zero

```
#include<iostream>
using namespace std;
int main()
{
int n;
cin>>n;
if(n>0)
cout<<"Positive";
else if(n<0)
cout<<"Negative";
else
cout<<"Zero";
return 0;
}
```

Output

15

Positive

TOPIC 5: DATA MEMBERS

1. Employee Annual Salary

```
#include<iostream>
using namespace std;
class Employee
{
public:
int empId;
string name;
float salary;
};
int main()
{
Employee e;
e.empId=101;
e.name="Rahul";
e.salary=30000;
cout<<"Annual Salary = "<<e.salary*12;
return 0;
}
```

Output

Annual Salary = 360000

---

2. Circle Area and Circumference

```
#include<iostream>
using namespace std;
class Circle
{
public:
float radius;
};
int main()
{
Circle c;
c.radius=7;
cout<<"Area = "<<3.14*c.radius*c.radius<<endl;
cout<<"Circumference = "<<2*3.14*c.radius;
return 0;
}
```

Output

Area = 153.86

Circumference = 43.96

---

3. Mobile Details

```
#include<iostream>
using namespace std;
class Mobile
{
public:
string brand;
string model;
int price;
int storage;
};
int main()
{
Mobile m;
```

```
m.brand="Samsung";
m.model="M35";
m.price=25000;
m.storage=128;
cout<<"Brand: "<<m.brand<<endl;
cout<<"Model: "<<m.model<<endl;
cout<<"Price: "<<m.price<<endl;
cout<<"Storage: "<<m.storage<<"GB";
return 0;
}
```

Output

Brand: Samsung

Model: M35

Price: 25000

Storage: 128GB

TOPIC 6: MEMBER FUNCTIONS

1. Calculator

```
#include<iostream>
using namespace std;
class Calculator
{
public:
void calculate(int a,int b)
{
cout<<"Addition = "<<a+b<<endl;
cout<<"Subtraction = "<<a-b<<endl;
cout<<"Multiplication = "<<a*b<<endl;
cout<<"Division = "<<a/b;
}
};
int main()
{
Calculator c;
c.calculate(20,10);
return 0;
}
```

Output

Addition = 30

Subtraction = 10

Multiplication = 200

Division = 2

---

2. Rectangle Area and Perimeter

```
#include<iostream>
using namespace std;
class Rectangle
{
public:
int l,b;
void display()
{
cout<<"Area = "<<l*b<<endl;
cout<<"Perimeter = "<<2*(l+b);
}
};
```

```
int main()
{
Rectangle r;
r.l=8;
r.b=4;
r.display();
return 0;
}
```

Output

Area = 32

Perimeter = 24

---

3. Prime Number

```
#include<iostream>
using namespace std;
class Number
{
public:
int n;
void check()
{
int count=0;
for(int i=1;i<=n;i++)
{
if(n%i==0)
count++;
}
if(count==2)
cout<<"Prime";
else
cout<<"Not Prime";
}
};
int main()
{
Number obj;
obj.n=13;
obj.check();
return 0;
}
```

Output

Prime

TOPIC 7: ARRAYS

1. Largest Element

```
#include<iostream>
using namespace std;
class Largest
{
public:
int a[5]={10,20,55,40,35};
void find()
{
int max=a[0];
for(int i=1;i<5;i++)
{
```

```
if(a[i]>max)
max=a[i];
}
cout<<"Largest = "<<max;
}
};
int main()
{
Largest l;
l.find();
return 0;
}
```

Output

Largest = 55

---

2. Sum and Average

```
#include<iostream>
using namespace std;
class Array
{
public:
int a[5]={10,20,30,40,50};
void calculate()
{
int sum=0;
for(int i=0;i<5;i++)
sum+=a[i];
cout<<"Sum = "<<sum<<endl;
cout<<"Average = "<<sum/5;
}
};
int main()
{
Array obj;
obj.calculate();
return 0;
}
```

Output

Sum = 150

Average = 30

---

3. Reverse Array

```
#include<iostream>
using namespace std;
class Reverse
{
public:
int a[5]={1,2,3,4,5};
void display()
{
for(int i=4;i>=0;i--)
cout<<a[i]<<" ";
}
};
int main()
{
```

```
Reverse r;
r.display();
return 0;
}
```

Output

5 4 3 2 1

TOPIC 8: FUNCTION INSIDE CLASS

1. Student Details

```
#include<iostream>
using namespace std;
class Student
{
public:
string name="Amit";
int roll=101;
void display()
{
cout<<"Name: "<<name<<endl;
cout<<"Roll No: "<<roll;
}
};
int main()
{
Student s;
s.display();
return 0;
}
```

Output

Name: Amit

Roll No: 101

---

2. Circle Area

```
#include<iostream>
using namespace std;
class Circle
{
public:
float r=5;
void area()
{
cout<<"Area = "<<3.14*r*r;
}
};
int main()
{
Circle c;
c.area();
return 0;
}
```

Output

Area = 78.5

---

3. Factorial

```
#include<iostream>
using namespace std;
```

```
class Number
{
public:
void factorial(int n)
{
int f=1;
for(int i=1;i<=n;i++)
f=f*i;
cout<<"Factorial = "<<f;
}
};
int main()
{
Number obj;
obj.factorial(5);
return 0;
}
```

Output

Factorial = 120

TOPIC 9: FUNCTION OUTSIDE THE CLASS

1. Student Display Function Outside Class

```
#include<iostream>
using namespace std;
class Student
{
public:
string name;
int roll;
void display();
};
void Student::display()
{
cout<<"Name: "<<name<<endl;
cout<<"Roll No: "<<roll;
}
int main()
{
Student s;
s.name="Amit";
s.roll=101;
s.display();
return 0;
}
```

Output

Name: Amit

Roll No: 101

---

2. Rectangle Area Function Outside Class

```
#include<iostream>
using namespace std;
class Rectangle
{
public:
```

```
int l,b;
int area();
};
int Rectangle::area()
{
return l*b;
}
int main()
{
Rectangle r;
r.l=10;
r.b=5;
cout<<"Area = "<<r.area();
return 0;
}
```

Output

Area = 50

---

3. Number Even or Odd

```
#include<iostream>
using namespace std;
class Number
{
public:
int n;
void check();
};
void Number::check()
{
if(n%2==0)
cout<<"Even";
else
cout<<"Odd";
}
int main()
{
Number obj;
obj.n=9;
obj.check();
return 0;
}
```

Output

Odd

TOPIC 10: ACCESS SPECIFIERS

1. Private Data and Public Function

```
#include<iostream>
using namespace std;
class Student
{
private:
int marks;
public:
void setMarks(int m)
{
marks=m;
```

```
}
void display()
{
cout<<"Marks = "<<marks;
}
};
int main()
{
Student s;
s.setMarks(85);
s.display();
return 0;
}
```

Output

```
Marks = 85
```

---

2. Public Data Members

```
#include<iostream>
using namespace std;
class Employee
{
public:
int id;
string name;
};
int main()
{
Employee e;
e.id=101;
e.name="Rahul";
cout<<e.id<<endl;
cout<<e.name;
return 0;
}
```

Output

```
101
Rahul
```

---

3. Protected Member

```
#include<iostream>
using namespace std;
class Base
{
protected:
int num;
};
class Derived:public Base
{
public:
void setValue()
{
num=100;
}
void display()
{
cout<<num;
```

```
}
};
int main()
{
Derived d;
d.setValue();
d.display();
return 0;
}
```

Output

100

TOPIC 11: DEFAULT CONSTRUCTOR

1. Student Default Constructor

```
#include<iostream>
using namespace std;
class Student
{
public:
string name;
int roll;
Student()
{
name="Amit";
roll=101;
}
void display()
{
cout<<name<<endl;
cout<<roll;
}
};
int main()
{
Student s;
s.display();
return 0;
}
```

Output

Amit

101

---

2. Rectangle Area

```
#include<iostream>
using namespace std;
class Rectangle
{
public:
int l,b;
Rectangle()
{
l=10;
b=5;
}
};
int main()
```

```
{
Rectangle r;
cout<<"Area = "<<r.l*r.b;
return 0;
}
```

Output

```
Area = 50
```

---

3. Circle Area

```
#include<iostream>
using namespace std;
class Circle
{
public:
int r;
Circle()
{
r=7;
}
};
int main()
{
Circle c;
cout<<"Area = "<<3.14c.rc.r;
return 0;
}
```

Output

```
Area = 153.86
```

TOPIC 12: PARAMETERIZED CONSTRUCTOR

1. Student Parameterized Constructor

```
#include<iostream>
using namespace std;
class Student
{
public:
int roll;
string name;
Student(int r,string n)
{
roll=r;
name=n;
}
void display()
{
cout<<roll<<endl;
cout<<name;
}
};
int main()
{
Student s(101,"Amit");
s.display();
return 0;
}
```

Output

101

Amit

---

2. Rectangle Area

```
#include<iostream>
using namespace std;
class Rectangle
{
public:
int l,b;
Rectangle(int x,int y)
{
l=x;
b=y;
}
};
int main()
{
Rectangle r(10,5);
cout<<"Area = "<<r.l*r.b;
return 0;
}
```

Output

Area = 50

---

3. Employee Constructor

```
#include<iostream>
using namespace std;
class Employee
{
public:
int id;
string name;
float salary;
Employee(int i,string n,float s)
{
id=i;
name=n;
salary=s;
}
void display()
{
cout<<id<<endl;
cout<<name<<endl;
cout<<salary;
}
};
int main()
{
Employee e(101,"Rahul",30000);
e.display();
return 0;
}
```

Output

101

Rahul

30000

TOPIC 13: SINGLE INHERITANCE

1. Person and Student

```
#include<iostream>
using namespace std;
class Person
{
public:
string name;
int age;
};
class Student:public Person
{
public:
int roll;
};
int main()
{
Student s;
s.name="Amit";
s.age=18;
s.roll=101;
cout<<s.name<<endl;
cout<<s.age<<endl;
cout<<s.roll;
return 0;
}
```

Output

Amit

18

101

---

2. Employee and Manager

```
#include<iostream>
using namespace std;
class Employee
{
public:
int id;
string name;
};
class Manager:public Employee
{
public:
string dept;
};
int main()
{
Manager m;
m.id=101;
m.name="Rahul";
m.dept="IT";
cout<<m.id<<endl;
cout<<m.name<<endl;
```

```
cout<<m.dept;
return 0;
}
```

Output

```
101
Rahul
IT
```

---

3. Animal and Dog

```
#include<iostream>
using namespace std;
class Animal
{
public:
void eat()
{
cout<<"Animal Eats"<<endl;
}
};
class Dog:public Animal
{
public:
void bark()
{
cout<<"Dog Barks";
}
};
int main()
{
Dog d;
d.eat();
d.bark();
return 0;
}
```

Output

```
Animal Eats
Dog Barks
```